EE3621: Digital Signal Processing

Lecture 7: The Discrete Fourier Transform (DFT) & Matrix Formulation
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins): Motivation & Introduction:** Why do we need the DFT? Transitioning from continuous-frequency DTFT to computable discrete-frequency DFT. The finite duration constraint.
- **08:00 – 15:00 (7 mins): Twiddle Factor Properties:** Mathematical definition, periodicity, and conjugate symmetry.
- **15:00 – 22:00 (7 mins): Orthogonality & IDFT Derivation:** Proving the orthogonality of complex exponentials and deriving the IDFT formula.
- **22:00 – 28:00 (6 mins): Matrix Formulation:** The $N \times N$ DFT matrix, explicit demonstration for $N = 4$, and DFT of common sequences.
- **28:00 – 34:00 (6 mins): Properties & Zero-Padding:** Key DFT properties table, spectral resolution, and zero-padding.
- **34:00 – 40:00 (6 mins): Worked Example & Checkpoints:** Step-by-step 4-point DFT calculation and conceptual checkpoint questions.

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2 Motivation: From DTFT to DFT

2.1 The Problem with the DTFT

Recall the Discrete-Time Fourier Transform (DTFT) for a sequence $x[n]$:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \tag{1}$$

While $x[n]$ is discrete in time, $X(e^{j\omega})$ is a **continuous function** of the frequency variable ω. In engineering, digital computers can only store and process finite, discrete sets of numbers. We cannot evaluate a continuous function over infinitely many points. Moreover, we observe signals for a **finite duration**.

2.2 The Solution: The DFT

To make frequency analysis computable, we must:

1. **Truncate** the signal to a finite number of samples, N.
2. **Sample** the continuous spectrum $X(e^{j\omega})$ at N equally spaced frequencies.

Evaluating the DTFT at $\omega_k = \frac{2\pi}{N}k$ for $k = 0, \ldots, N-1$ gives the **Discrete Fourier Transform (DFT)**. This maps a finite N-point time sequence to an N-point frequency sequence.

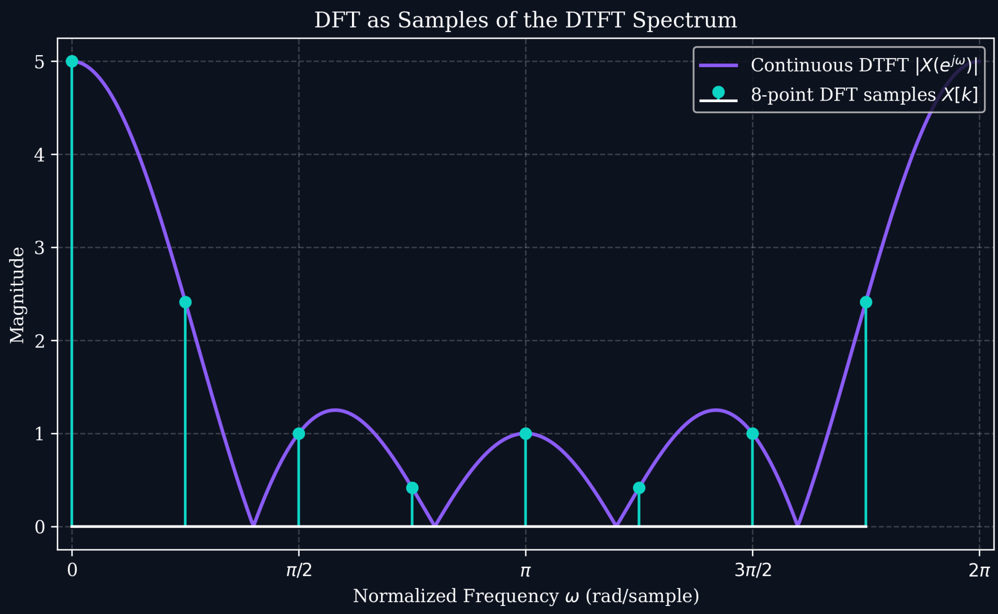

Figure 1: Sampling the continuous DTFT spectrum to obtain discrete DFT bins.

3 The Twiddle Factor and Its Properties

To simplify notation, we introduce the **Twiddle Factor**:

$$W_N = e^{-j\frac{2\pi}{N}} \tag{2}$$

The N-point DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \le k \le N-1 \tag{3}$$

3.1 Properties of the Twiddle Factor

1. **Exponentiation to N:**

$$W_N^N = \left(e^{-j\frac{2\pi}{N}}\right)^N = e^{-j2\pi} = 1 \implies W_N^{kN} = 1^k = 1$$

2. **Periodicity:**

$$W_N^{k+N} = W_N^k \cdot W_N^N = W_N^k \cdot 1 = W_N^k$$

3. **Conjugate Symmetry:**

$$W_N^{k(N-n)} = W_N^{kN} \cdot W_N^{-kn} = 1 \cdot (W_N^{kn})^{-1} = e^{j\frac{2\pi}{N}kn} = (W_N^{kn})^*$$

4 Orthogonality Proof and IDFT Derivation

4.1 Orthogonality Proof

Evaluate the sum of the product of two complex exponentials:

$$S = \sum_{n=0}^{N-1} W_N^{kn} W_N^{-ln} = \sum_{n=0}^{N-1} W_N^{(k-l)n} \tag{4}$$

Let $m = k - l$. This is a geometric series $\sum_{n=0}^{N-1} (W_N^m)^n$.

Case 1: $k = l$ (or $m = 0$)

The ratio is $W_N^0 = 1$.

$$S = \sum_{n=0}^{N-1} 1^n = N$$

Case 2: $k \neq l$

The geometric series sum formula gives:

$$S = \frac{1 - (W_N^m)^N}{1 - W_N^m} = \frac{1 - W_N^{mN}}{1 - W_N^m} = \frac{1 - 1}{1 - W_N^m} = 0$$

KEY RESULT: Orthogonality Condition

$$\sum_{n=0}^{N-1} W_N^{kn} W_N^{-ln} = N\delta[k - l] \tag{5}$$

4.2 Derivation of the IDFT

Starting with the DFT formula:

$$X[k] = \sum_{m=0}^{N-1} x[m] W_N^{km}$$

Multiply both sides by W_N^{-kn} and sum over k:

$$\sum_{k=0}^{N-1} X[k] W_N^{-kn} = \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] W_N^{km} \right) W_N^{-kn}$$

Interchanging summation orders:

$$\sum_{k=0}^{N-1} X[k] W_N^{-kn} = \sum_{m=0}^{N-1} x[m] \left(\sum_{k=0}^{N-1} W_N^{k(m-n)} \right)$$

By orthogonality, the inner sum is $N\delta[m - n]$:

$$\sum_{k=0}^{N-1} X[k] W_N^{-kn} = x[n] \cdot N$$

Dividing by N gives the Inverse Discrete Fourier Transform (IDFT):

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq n \leq N - 1 \tag{6}$$

5 Matrix Formulation of the DFT

The DFT can be expressed as a matrix multiplication $\mathbf{X} = \mathbf{W}_N \mathbf{x}$:

$$\mathbf{W}_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^1 & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix} \tag{7}$$

For $N = 4$, with $W_4 = -j$:

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \tag{8}$$

The IDFT in matrix form is $\mathbf{x} = \frac{1}{N}\mathbf{W}_N^* \mathbf{X}$.

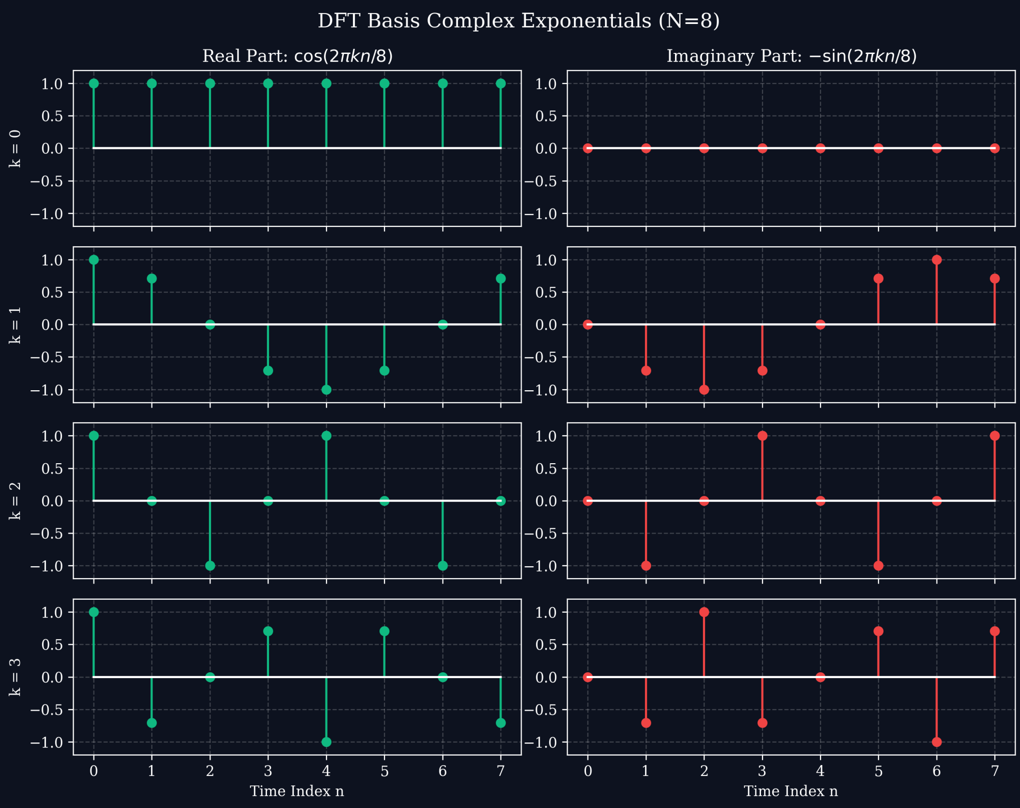

Figure 2: Real and Imaginary basis functions for N=8.

6 DFT of Common Sequences

- **Unit Impulse** $x[n] = \delta[n] \implies X[k] = 1$ (Flat spectrum)
- **DC Sequence** $x[n] = 1 \implies X[k] = N\delta[k]$ (Zero-frequency spike)
- **Complex Exponential** $x[n] = e^{j\frac{2\pi}{N}mn} \implies X[k] = N\delta[k-m]$

7 Spectral Resolution and Zero-Padding

The frequency resolution is $\Delta\omega = \frac{2\pi}{N}$ or $\Delta f = \frac{f_s}{N}$. To improve the *display resolution* (making the spectrum appear smoother and denser), we can append zeros to the signal, increasing the DFT size N. Zero-padding interpolates the DTFT spectrum but does not improve the fundamental ability to resolve closely spaced frequencies.

8 DFT Properties Table

Property	**Time Domain** $x[n]$	**Frequency Domain** $X[k]$
Linearity	$ax_1[n] + bx_2[n]$	$aX_1[k] + bX_2[k]$
Circular Time Shift	$x[(n-m) \pmod N]$	$X[k]W_N^{km}$
Circular Freq Shift	$x[n]W_N^{-ln}$	$X[(k-l) \pmod N]$
Circular Convolution	$x_1[n] \circledast x_2[n]$	$X_1[k]X_2[k]$
Conjugate Symmetry	If $x[n]$ is real	$X[k] = X^*[N-k]$

9 Worked Example: Step-by-Step DFT Computation

Find the 4-point DFT of $x[n] = \{1, 1, 0, 0\}$.

$$\mathbf{X} = \mathbf{W}_4\mathbf{x} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} X[0] &= 1(1) + 1(1) + 1(0) + 1(0) = 2 \\ X[1] &= 1(1) + (-j)(1) + (-1)(0) + j(0) = 1 - j \\ X[2] &= 1(1) + (-1)(1) + 1(0) + (-1)(0) = 0 \\ X[3] &= 1(1) + j(1) + (-1)(0) + (-j)(0) = 1 + j \end{aligned}$$

Result: $X[k] = \{2, 1-j, 0, 1+j\}$.

10 Checkpoint Questions

1. **Prove that the 4-point DFT of $x[n] = \{1, -1, 1, -1\}$ is $X[k] = \{0, 0, 4, 0\}$.**
   Ans: The sequence is $\cos(\pi n) = (-1)^n = e^{j\pi n} = W_4^{-2n}$. This perfectly aligns with the $k = 2$ basis vector. By orthogonality, the inner product yields $N = 4$ at $k = 2$ and 0 elsewhere.

2. **For a 1-second audio signal at 8000 Hz, how do you get 0.5 Hz DFT resolution?**
   Ans: $\Delta f = f_s/N \implies 0.5 = 8000/N \implies N = 16000$. The signal has 8000 samples, so you must zero-pad by appending 8000 zeros.

3. **If $x[n]$ is real and 8-point DFT $X[1] = 3 + j2$, what is $X[7]$?**
   Ans: $X[7] = 3 - j2$. By conjugate symmetry for real signals, $X[k] = X^*[N-k]$, so $X[7] = X^*[1]$.